

# Penetration Test Report

AI-Powered Windows Security Assessment

HIGH RISK

|               |                      |
|---------------|----------------------|
| Target IP     | 172.20.10.5          |
| Hostname      | N/A                  |
| OS Detected   | Windows              |
| Scan Date     | May 17, 2026 — 10:14 |
| Duration      | 12.24 seconds        |
| Open Services | 3                    |
| CVEs Matched  | 12                   |
| Risk Score    | 6.6 / 10 — HIGH RISK |

This report was generated by the AI-Powered Windows Penetration Testing Assistant. For authorized security assessment use only.

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## 01 — Executive Summary

This assessment identified **3 open service(s)** on the target system and matched **12 CVE vulnerabilities** from the local Windows CVE database. Overall risk: **HIGH RISK**. Risk score: **6.6/10**.

Of the 12 matched CVEs, 2 are recent (post-2020) and may be unpatched, while 10 are older CVEs likely patched on modern Windows systems.

| Severity | Count | Priority                     |
|----------|-------|------------------------------|
| CRITICAL | 2     | Immediate action required    |
| HIGH     | 7     | Remediate within 24-48 hours |
| MEDIUM   | 3     | Remediate within 30 days     |
| LOW      | 0     | Remediate when possible      |

### AI Assessment Overview

Executive Summary Overall Risk Assessment: The target, a Windows system with open ports for RPC and SMB services, is highly vulnerable due to multiple critical vulnerabilities across both protocols. Specifically, the 135/RPC port contains three high-severity vulnerabilities (CVE-2020-1113, CVE-2017...

## 02 — Scan Information

|             |                                                                                     |
|-------------|-------------------------------------------------------------------------------------|
| Target IP   | 172.20.10.5                                                                         |
| Hostname    | N/A                                                                                 |
| OS Detected | Windows                                                                             |
| Scan Date   | 2026-05-17 10:14:38                                                                 |
| Duration    | 12.24 seconds                                                                       |
| Scanner     | Nmap 7.98                                                                           |
| Command     | /usr/lib/nmap/nmap --privileged -vvv -p 135,139,445 -4 -sV -sC --script smb-vuln... |

## 03 — Open Ports & Services

| Port | Protocol | Service | Product / Banner              |
|------|----------|---------|-------------------------------|
| 135  | TCP      | RPC     | Microsoft Windows RPC         |
| 139  | TCP      | SMB     | Microsoft Windows netbios-ssn |
| 445  | TCP      | SMB     | microsoft-ds?                 |

## 04 — CVE Vulnerability Findings

CVE matches are based on service detection from the local NIST NVD dataset. Patch status is estimated by publication date.

### Service: 135/rpc

| CVE ID        | CVSS | Severity | Description                                                                                   | Patch Status   |
|---------------|------|----------|-----------------------------------------------------------------------------------------------|----------------|
| CVE-2016-0178 | 8.8  | HIGH     | The RPC NDR Engine in Microsoft Windows Vista SP2, Windows Server 2008 SP2 and R2 SP1, Win... | Likely Patched |

|                      |     |               |                                                                                               |                         |
|----------------------|-----|---------------|-----------------------------------------------------------------------------------------------|-------------------------|
| <b>CVE-2017-0100</b> | 7.8 | <b>HIGH</b>   | A DCOM object in Helppane.exe in Microsoft Windows 7 SP1; Windows Server 2008 R2; Windows ... | Likely Patched          |
| <b>CVE-2020-1113</b> | 7.5 | <b>HIGH</b>   | A security feature bypass vulnerability exists in Microsoft Windows when the Task Schedule... | <b>May be Unpatched</b> |
| <b>CVE-2016-0128</b> | 6.8 | <b>MEDIUM</b> | The SAM and LSAD protocol implementations in Microsoft Windows Vista SP2, Windows Server 2... | Likely Patched          |

#### Service: 139/smb

| <b>CVE ID</b>         | <b>CVSS</b> | <b>Severity</b> | <b>Description</b>                                                                            | <b>Patch Status</b>     |
|-----------------------|-------------|-----------------|-----------------------------------------------------------------------------------------------|-------------------------|
| <b>CVE-2017-0021</b>  | 9.0         | <b>CRITICAL</b> | Hyper-V in Microsoft Windows 10 1607 and Windows Server 2016 does not properly validate vS... | Likely Patched          |
| <b>CVE-2015-2474</b>  | 9.0         | <b>CRITICAL</b> | Microsoft Windows Vista SP2 and Server 2008 SP2 allow remote authenticated users to execut... | Likely Patched          |
| <b>CVE-2017-0144</b>  | 8.8         | <b>HIGH</b>     | The SMBv1 server in Microsoft Windows Vista SP2; Windows Server 2008 SP2 and R2 SP1; Windo... | Likely Patched          |
| <b>CVE-2017-0145</b>  | 8.8         | <b>HIGH</b>     | The SMBv1 server in Microsoft Windows Vista SP2; Windows Server 2008 SP2 and R2 SP1; Windo... | Likely Patched          |
| <b>CVE-2022-24500</b> | 8.8         | <b>HIGH</b>     | Windows SMB Remote Code Execution Vulnerability                                               | <b>May be Unpatched</b> |
| <b>CVE-2017-0146</b>  | 8.8         | <b>HIGH</b>     | The SMBv1 server in Microsoft Windows Vista SP2; Windows Server 2008 SP2 and R2 SP1; Windo... | Likely Patched          |
| <b>CVE-2018-8444</b>  | 5.9         | <b>MEDIUM</b>   | An information disclosure vulnerability exists in the way that the Microsoft Server Messag... | Likely Patched          |
| <b>CVE-2017-0276</b>  | 5.9         | <b>MEDIUM</b>   | Microsoft Server Message Block 1.0 (SMBv1) allows an information disclosure vulnerability ... | Likely Patched          |

# 05 — AI Analysis & Recommendations

Analysis generated by Qwen2.5:3b AI model using RAG from local CVE dataset.

## Executive Summary

Overall Risk Assessment: The target, a Windows system with open ports for RPC and SMB services, is highly vulnerable due to multiple critical vulnerabilities across both protocols. Specifically, the 135/RPC port contains three high-severity vulnerabilities (CVE-2020-1113, CVE-2017-0144, CVE-2016-0128), and the 139/SMB port has eight critical vulnerabilities (CVE-2017-0021, CVE-2015-2474, CVE-2022-

## Most Critical Vulnerabilities Identified:

- CVE-2020-1113 (Task Scheduler Security Feature Bypass)
- CVE-2017-0021 (Hyper-V System Data Structure Corruption)

## Vulnerability Analysis

Rank by Exploitability and Impact

CVE ID | Severity | Published | Patch Status

CVE-2020-1113 | HIGH | 2020-05-21 | ✓ Recent CVE — may still be unpatched

CVE-2017-0021 | CRITICAL | 2017-03-17 | ■■■ Likely patched on modern Windows 10/11 (published pre-2020)

## CVE-2020-1113:

This vulnerability allows attackers to execute arbitrary code on a target OS, making it highly critical. The Task Scheduler service is a core part of the operating system and can be used for remote code execution.

## CVE-2017-0021:

The Hyper-V server in Windows 1607 and Server 2016 does not properly validate vSMB packet data, which allows attackers to execute arbitrary code. This is a critical vulnerability that impacts the virtualization layer of the operating system.

## Attack Vectors

Most Realistic Attack Paths

## CVE-2020-1113 (Task Scheduler Security Feature Bypass):

1. Exploit Method: The attacker would exploit this vulnerability by sending crafted packets to a vulnerable Task Scheduler service.
2. Metasploit Module: msfvenom can be used to generate the malicious payload, and msfconsole for exploitation.

## 3. Steps:

- Identify the vulnerable Task Scheduler service using tools like Nmap or similar scanning software.
- Craft a malicious payload that exploits CVE-2020-1113.
- Execute the payload on the target system.

## CVE-2017-0021 (Hyper-V System Data Structure Corruption):

1. Exploit Method: The attacker would send crafted packets to the SMBv1 server, which is vulnerable due to this CVE.
2. Metasploit Module: Use msfconsole for exploitation and Metasploit modules like msfvenom for crafting payloads.

## 3. Steps:

- Identify the SMBv1 server using tools like Nmap or similar scanning software.
- Craft a malicious payload that exploits CVE-2017-0021.
- Execute the payload on the target system.

Recommended Tools & Techniques

Metasploit Modules

CVE ID | Metasploit Module

CVE-2020-1113 | exploit/windows/smb/ms16\_058\_sam

CVE-2017-0021 | exploit/multi/handler (for testing)

Other Tools

- Nmap: For scanning and identifying vulnerable services.
- Burp Suite: For intercepting and analyzing SMB traffic.

Remediation Steps

CVE-2020-1113:

- Apply the latest security patches provided by Microsoft for Windows 10/11. Ensure that all relevant updates are installed to mitigate this vulnerability.

CVE-2017-0021:

- Upgrade to a more recent version of Hyper-V, such as Windows Server 2019 or later, which includes the necessary security patches.
- Alternatively, apply the Microsoft Security Update KB4538465 for Windows Server 2016.

Risk Score

Overall Score: 8/10

Justification: The target is highly vulnerable to multiple critical vulnerabilities. CVE-2020-1113 and CVE-2017-0021 are the most severe, with CVE-2020-1113 having a CVSS score of 8.8 and CVE-2017-0021 having a CVSS score of 9.0. These vulnerabilities have significant real-world exploitability and impact.

Conclusion

The target system is highly vulnerable to multiple critical vulnerabilities, including the Task Scheduler Security Feature Bypass (CVE-2020-1113) and Hyper-V System Data Structure Corruption (CVE-2017-0021). Immediate remediation should be prioritized to mitigate these risks.

06 — Remediation Steps

|     |                                                                                                 |
|-----|-------------------------------------------------------------------------------------------------|
| RPC | Restrict RPC via firewall. Apply all Windows security patches. Disable unused RPC services.     |
| SMB | Disable SMBv1. Apply MS17-010 patch (KB4012212). Block port 445 externally. Enable SMB signing. |

General Security Hardening

- Apply all pending Windows security updates via Windows Update
- Enable Windows Defender and keep definitions up to date
- Implement network segmentation and strict firewall rules

- Use principle of least privilege for all accounts
- Enable Windows Event Logging and monitor for suspicious activity
- Implement multi-factor authentication (MFA) for all remote access

## 07 — Risk Score

|                    |           |
|--------------------|-----------|
| Overall Risk Score | 6.6 / 10  |
| Risk Level         | HIGH RISK |
| Total CVEs         | 12        |
| Critical           | 2         |
| High               | 7         |
| Medium             | 3         |
| Low                | 0         |
| Recent CVEs        | 2         |
| Likely Patched     | 10        |

Risk score is calculated based on recent CVEs (post-2020) only. Older CVEs are weighted lower as they are likely patched on modern Windows systems.

## 08 — Legal Disclaimer

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This report was prepared exclusively for the authorized recipient. All testing was conducted with explicit written authorization from the system owner.

Unauthorized access to computer systems is a criminal offence under the Computer Misuse Act 1990 (UK), Computer Fraud and Abuse Act (USA), and equivalent legislation worldwide.

CVE data sourced from the National Vulnerability Database (NVD) — NIST (<https://nvd.nist.gov/>). AI analysis by Qwen2.5:3b via Ollama (RAG mode — local dataset only).

Report generated: 2026-05-17 10:14:38